Vivekanand Education Society's
Institute of Technology

(Affiliated to University of Mumbai, Approved by AICTE & Recognized by Govt. of Maharashtra)

Title

XAI on Music Recommender System

Submitted in partial fulfilment of the requirements

of the degree of

Bachelor of Engineering in

Artificial Intelligence and Data Science

by

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Vivekanand Education Society's Institute of Technology

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Department of Artificial Intelligence and Data Science

CERTIFICATE

This is to certify that **Anjali Parwani, Abhishek Pattanayak, Divyam Poptani** and **Rashmit Vartak** of Second Year of Artificial Intelligence and Data Science studying under the University of Mumbai have satisfactorily presented the Mini Project entitled XAI on **Music Recommender System** as a part of the MINI-PROJECT for Semester-IV under the guidance of **Mrs. Sangeeta Oswal** in the year 2022-2023.

Date:

(Name and sign)	**(Name and sign)**	**(Name and sign)**
Head of Department	**Supervisor/Guide**	**Examiner**

Department of Artificial Intelligence and Data Science

DECLARATION

We, **Anjali Parwani, Abhishek Pattanayak, Divyam Poptani** and **Rashmit Vartak** from **D6AD**, declare that this project represents our ideas in our own words without plagiarism and wherever others' ideas or words have been included, we have adequately cited and referenced the original sources.

We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our project work.

We declare that we have maintained a minimum 75% attendance, as per the University of Mumbai norms.

We understand that any violation of the above will be cause for disciplinary action by the Institute.

Yours Faithfully

1. Anjali Parwani
2. Abhishek Pattanayak
3. Divyam Poptani
4. Rashmit Vartak

(Name & Signature of Students with Date)

Acknowledgement

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ABSTRACT

Explaining the recommendations provided by music recommender systems can be a challenging task, as it requires interpreting complex algorithms and data processing techniques. Explainable Artificial Intelligence (XAI) provides a solution to this problem by providing insights into the decision-making process of these systems.

XAI techniques such as SHaP(SHapely addictive Prediction) and Local Interpretable Model-agnostic Explanations (LIME) have been applied to music recommender systems to provide transparency and accountability. These techniques enable users to understand how the system generates recommendations and how it takes into account various factors such as user preferences, previous listening history, and contextual information.

XAI on music recommender systems has the potential to enhance user satisfaction and trust, increase system usability, and facilitate the discovery of new and diverse music. However, it also poses challenges such as the trade-off between interpretability and accuracy, and the need for effective visualization techniques to present the explanations to users in a meaningful way.

Overall, XAI on music recommender systems is a promising area of research that can improve the transparency and interpretability of these systems and ultimately enhance the user experience.

LIST OF FIGURES

1.INTRODUCTION

1.1 INTRODUCTION

The Music Recommender System is a popular application of machine learning that provides personalized music recommendations to users based on their listening history, preferences, and behavior. However, traditional recommendation systems are often criticized for their lack of transparency and interpretability, making it difficult for users to understand why certain recommendations are made. To address this issue, we propose an XAI (Explainable Artificial Intelligence) approach to the Music Recommender System that provides explanations for each recommendation, helping users to understand and trust the system's recommendations.

In this project, we will develop an XAI-powered Music Recommender System that provides explanations for each recommendation made. We will explore various XAI techniques such as LIME, SHAP, and attention mechanisms to provide local and global explanations for each recommendation. We will evaluate the performance of the XAI-powered Recommender System against traditional recommendation systems using metrics such as accuracy, coverage, and diversity. We will also conduct user studies to evaluate the effectiveness and user acceptance of the XAI explanations.

The results of this project can have significant implications for the music industry, helping users to discover new music and providing a more transparent and trustworthy experience for music recommendation systems. The techniques developed in this project can also be applied to other recommendation systems in domains such as e-commerce, healthcare, and finance, where XAI-powered systems can improve user trust and satisfaction.

1.2 PROBLEM STATEMENT

Traditional Music Recommender Systems are popular for providing personalized music recommendations to users based on their listening history, preferences, and behavior. However, these systems lack transparency and interpretability, making it difficult for users to understand why certain recommendations are made. This lack of transparency and interpretability can lead to a lack of user trust and adoption of the system.

To address this issue, we propose an XAI-powered Music Recommender System that provides explanations for each recommendation made. The aim of this project is to develop an XAI-powered Recommender System that can provide personalized music recommendations to users while also providing clear and transparent explanations for each recommendation made.

1.3 OBJECTIVES

The project's objectives are:

1. Develop an XAI-powered Music Recommender System that provides explanations for each recommendation made.
2. Explore various XAI techniques such as LIME, SHAP, and attention mechanisms to provide local and global explanations for each recommendation.
3. Evaluate the performance of the XAI-powered Recommender System against traditional recommendation systems using metrics such as accuracy, coverage, and diversity.
4. Conduct user studies to evaluate the effectiveness and user acceptance of the XAI explanations.

1.4 SCOPE

The outcomes of this project will provide insights into the effectiveness and usability of XAI techniques for Music Recommender Systems. The results can also have significant implications for other recommendation systems, improving user trust, and satisfaction in domains such as e-commerce, healthcare, and finance.

2. LITERATURE SURVEY

2.1 LITERATURE/ TECHNIQUES STUDIED

Some of the literature and techniques that can be studied and applied in the XAI-powered Music Recommender System project are:

1. **Collaborative Filtering:** Collaborative filtering is a traditional recommendation system technique that recommends items to users based on their past behavior and the behavior of similar users. It is a widely used approach in the music recommendation domain and can be used as a baseline for comparison with the XAI-powered Recommender System.
2. **Content-based Filtering:** Content-based filtering is another traditional recommendation system technique that recommends items to users based on their preferences and characteristics of the items. In the music domain, it involves analyzing music features such as genre, tempo, and artist to make recommendations.
3. **K- Means:** It is an unsupervised learning technique that aims to group similar data points together into clusters. The algorithm starts by randomly selecting K centroids (where K is a user-defined parameter), and then assigns each data point to the nearest centroid. The mean of each cluster is then calculated and used as the new centroid for that cluster. This process is repeated until the centroids no longer move significantly, or a maximum number of iterations is reached.
4. **KNN(clustering):** K-nearest neighbors (KNN) uses the training data to make predictions based on the K closest data points to the query point (where K is a user-defined parameter).In the classification problem, it predicts the class label of the query point based on the majority class of the K nearest neighbors. It is also robust to noisy data and can adapt well to complex decision boundaries.
5. **DB Scan:** Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is an unsupervised learning technique that can discover clusters of arbitrary shapes in data and is robust to noise and outliers. The algorithm assigns each point to a cluster based on its connectivity to the core points.

The clusters can have arbitrary shapes and sizes and are separated by regions of low-density points.

6. **Attention Mechanisms:** Attention mechanisms are a class of XAI techniques that provide interpretability to deep learning models. They assign weights to the input features based on their importance to the model's predictions, allowing for easy interpretation and explanation.
7. **Hybrid Approaches:** Hybrid approaches combine multiple recommendation techniques to improve the accuracy and diversity of recommendations. They can be used to combine the strengths of collaborative filtering, content-based filtering, and XAI-powered approaches to create a more effective Music Recommender System.
8. **XAI:** Explainable Artificial Intelligence (XAI) is an emerging field of research that aims to create AI systems that can explain their decisions and actions in a transparent and interpretable way. XAI techniques enable users to understand how an AI system generates its output and what factors it considers in the decision-making process.There are several XAI techniques, such as decision trees, rule-based systems, Local Interpretable Model-agnostic Explanations (LIME),SHapley Additive exPlanations (SHAP) and attention mechanisms, among others.
9. **Local Interpretable Model-Agnostic Explanations** (LIME): **LIME** is an XAI technique that provides interpretable explanations for individual predictions made by a machine learning model. It generates local linear approximations of the model's predictions for a given input, allowing for easy interpretation of feature importance.
10. **SHapley Additive exPlanations**(SHAP): **SHAP** is an XAI technique that provides explanations for individual predictions based on the Shapley values from cooperative game theory. It measures the contribution of each feature to a given prediction, allowing for global and local interpretability of the model.

These literature and techniques can be studied and applied in the XAI-powered Music Recommender System project to develop an effective and transparent recommendation system that provides clear and interpretable explanations for each recommendation made.

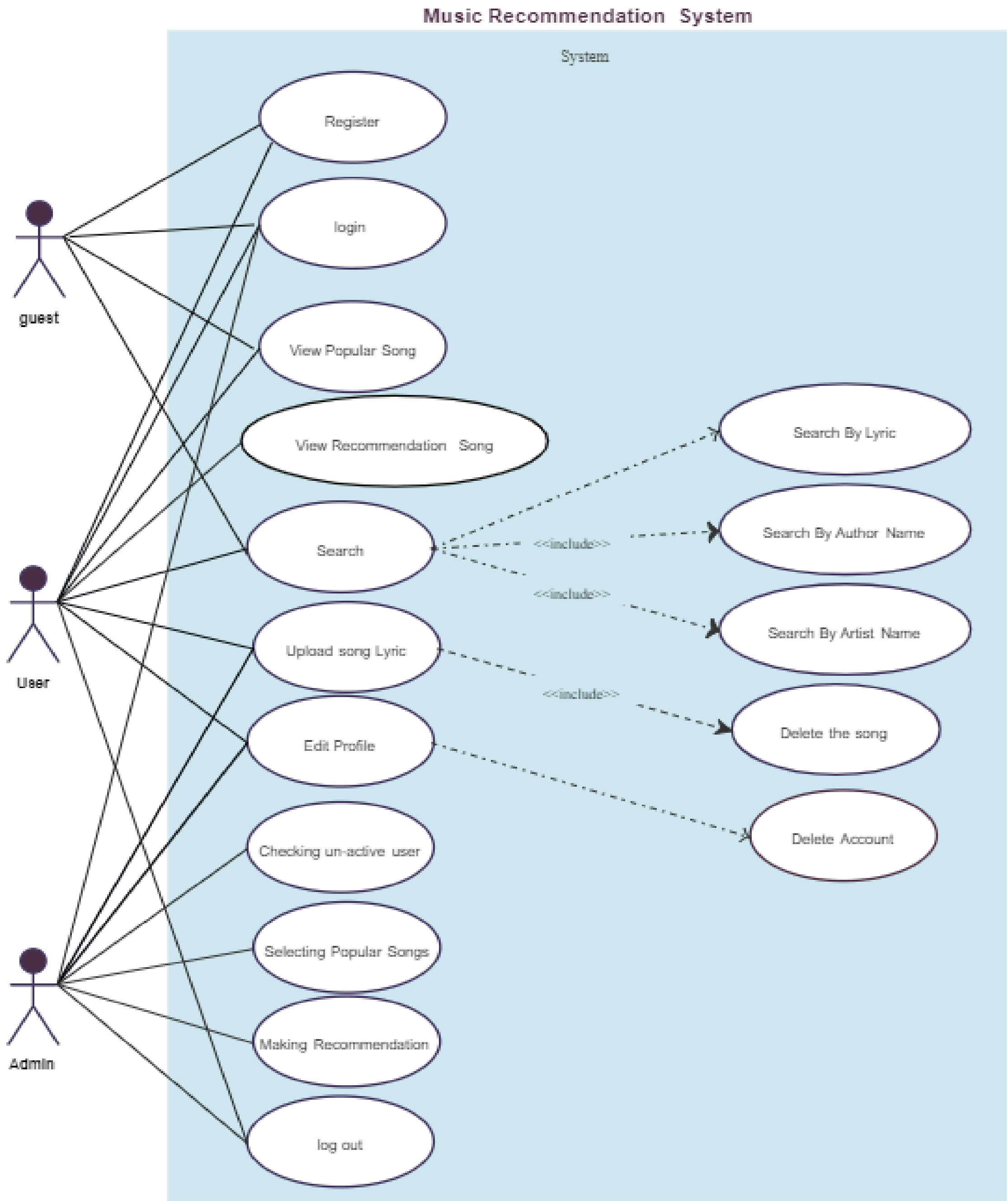

Fig 2.1: Basic working of a Music Recommendation System

2.2 PAPERS/FINDINGS (RESEARCH PAPERS / WHITE PAPERS / EXISTING SYSTEMS)

Here are some relevant research papers, white papers, and existing systems related to XAI-powered Music Recommender Systems:

1. "Explainable Recommendation: A Survey and New Perspectives" (2019) by Yuanfu Lu, Xinyi Zhang, et al: This research paper provides an overview of XAI techniques used in recommendation systems, including LIME, SHAP, and attention mechanisms. It also discusses the importance of interpretability in recommendation systems and provides case studies of XAI-powered recommendation systems in different domains.

2. "Explainable Artificial Intelligence for Music Understanding" (2019) by Simonetta Balsamo, et al. : This research paper proposes an XAI-powered system for music understanding that can provide explanations for music analysis and classification tasks. The system uses LIME and SHAP to provide explanations for individual predictions made by a deep learning model.

3. "Towards Explainable Music Mood Classification Using Attention-based Neural Networks" (2018) by Paria Yousefi, et al. : This research paper proposes an XAI-powered system for music mood classification that uses attention mechanisms to provide explanations for the model's predictions. The system was evaluated on a dataset of music tracks annotated with mood labels.

4. Spotify's Recommendation System: Spotify's music recommendation system is a widely used and successful music recommendation system that uses collaborative filtering and content-based filtering techniques. However, Spotify has also been exploring XAI techniques to improve the transparency and interpretability of its recommendation system.

5. Pandora's Music Genome Project: Pandora's Music Genome Project is a music recommendation system that uses a content-based filtering approach. The system analyzes the musical characteristics of songs such as melody, harmony, and rhythm to make recommendations.

These research papers and existing systems can provide valuable insights and inspiration for the development of the XAI-powered Music Recommender System project. They can also be used as baselines for comparison with the proposed XAI-powered system.

3. DATA SET

3.1 DESCRIPTION OF DATA SET

The dataset is intended to be used for building music recommendation systems and contains two files: data.csv and data_by_artist.csv.

data.csv includes information about the songs, including their name, artist, release date, and audio features such as danceability, energy, loudness, and tempo. data_by_artist.csv includes aggregated information about the artists, including their name, genres, and average audio features for their songs.

The dataset can be used to train machine learning models to make music recommendations based on user preferences and listening history.

3.2 DATA COLLECTION METHODOLOGY

Data scraping involves using software tools to extract information from websites or other digital sources. In the case of a music recommendation system, data scraping might involve using the Spotify API or other music-related websites to gather information about songs, artists, genres, and other relevant data points.

For example, the dataset available on Kaggle may have been created by scraping data from the Spotify API, which allows developers to access information about songs, albums, artists, and other music-related information. By using the Spotify API, the creator of the dataset may have been able to collect large amounts of data about songs and artists, including audio features such as tempo, energy, and danceability.

1	valence	year	acousticne	artists	danceabili	duration_r	energy	explicit	id	instrument	key	liveness	loudness	mode	name	popularity	release_da	speechine	tempo
2	0.0594	1921	0.982	['Sergei Ra	0.279	831667	0.211	0	4BJqT0PrA	0.878	10	0.665	-20.096	1	Piano Con	4	1921	0.0366	80.954
3	0.963	1921	0.732	['Dennis Da	0.819	180533	0.341	0	7xPhfUan2	0	7	0.16	-12.441	1	Clancy Lov	5	1921	0.415	60.936
4	0.0394	1921	0.961	['KHP Kridh	0.328	500062	0.166	0	1o6I8BglA	0.913	3	0.101	-14.85	1	Gati Bali	5	1921	0.0339	110.339
5	0.165	1921	0.967	['Frank Par	0.275	210000	0.309	0	3ftBPsC5vI	2.77E-05	5	0.381	-9.316	1	Danny Boy	3	1921	0.0354	100.109
6	0.253	1921	0.957	['Phil Rega	0.418	166693	0.193	0	4d6HGyGT	1.68E-06	3	0.229	-10.096	1	When Irish	2	1921	0.038	101.665
7	0.196	1921	0.579	['KHP Kridh	0.697	395076	0.346	0	4pyw9DVH	0.168	2	0.13	-12.506	1	Gati Mardi	6	1921	0.07	119.824
8	0.406	1921	0.996	['John McC	0.518	159507	0.203	0	5uNZnElqC	0	0	0.115	-10.589	1	The Weari	4	1921	0.0615	66.221
9	0.0731	1921	0.993	['Sergei Ra	0.389	218773	0.088	0	02GDntOX	0.527	1	0.363	-21.091	0	Morceaux	2	1921	0.0456	92.867
10	0.721	1921	0.996	['Ignacio C	0.485	161520	0.13	0	05xDjWH9	0.151	5	0.104	-21.508	0	La MaÃ±ar	0	########	0.0483	64.678
11	0.771	1921	0.982	['FortugÃ©	0.684	196560	0.257	0	08zfJvRLp7	0	8	0.504	-16.415	1	Il Etait Syn	0	1921	0.399	109.378
12	0.826	1921	0.995	['Maurice	0.463	147133	0.26	0	0BMkRpQt	0	9	0.258	-16.894	1	Dans La Vi	0	1921	0.0557	85.146
13	0.578	1921	0.994	['Ignacio C	0.378	155413	0.115	0	0F30WM8	0.906	10	0.11	-27.039	0	Por Que M	0	########	0.0414	70.37
14	0.493	1921	0.99	['Georgel']	0.315	190800	0.363	0	0H3k2CvJv	0	5	0.292	-12.562	0	La VipÃ¨re	0	1921	0.0546	174.532
15	0.212	1921	0.912	['Mehmet	0.415	184973	0.42	0	0LcXzABeA	0.89	8	0.108	-10.766	0	Ud Taksim	0	1921	0.114	70.758
16	0.493	1921	0.0175	['Zay Gatsb	0.527	205072	0.691	1	0MJZ4hh6	0.384	7	0.358	-7.298	1	Power Is P	0	########	0.0326	159.935
17	0.282	1921	0.989	['Sergei Ra	0.384	221013	0.171	0	0NFeJgmT	0.82	7	0.116	-20.476	0	10 PrÃ©lu	4	1921	0.0319	107.698
18	0.218	1921	0.957	['Phil Rega	0.259	186467	0.212	0	0Nk5f07H	0.000222	2	0.236	-13.3	1	Come Bacl	1	1921	0.0358	85.726
19	0.664	1921	0.996	['Hector B	0.541	250747	0.283	0	0POO8XaU	0.898	9	0.393	-14.808	1	RÃ¡kÃ³czy	0	1921	0.0477	108.986
20	0.0778	1921	0.148	['THE GUY	0.604	204957	0.418	1	0QQmUf4	0.0382	4	0.102	-11.566	0	When We	0	########	0.0417	80.073
21	0.527	1921	0.971	['Christoph	0.54	122000	0.0848	0	0QU5xT6N	0.00196	5	0.0887	-16.055	0	A Ballynure	0	1921	0.075	100.296
22	0.672	1921	0.994	['FortugÃ©	0.67	191333	0.113	0	0QvUUTHY	0	10	0.213	-16.57	0	Je Suis Tou	0	1921	0.196	87.162
23	0.24	1921	0.994	['John McC	0.4	187333	0.155	0	0RPKAq5yl	4.33E-05	4	0.103	-13.976	1	Mother Ma	0	1921	0.0873	170.251
24	0.422	1921	0.995	['Ignacio C	0.648	154240	0.0995	0	0SK1upzAF	0.846	11	0.112	-22.429	1	Flor March	0	########	0.105	71.978
25	0.381	1921	0.995	['Hanende	0.223	158908	0.245	0	0UqiUmGD	0.876	7	0.354	-14.387	1	Ä°mtidadÄ	0	########	0.0387	142.136
26	0.41	1921	0.97	['Mehmet	0.269	86988	0.143	0	0VhIFYGSp	0.469	4	0.166	-9.003	1	Ney Taksir	0	1921	0.0413	141.386
27	0.0731	1921	0.993	['Sergei Ra	0.389	218773	0.088	0	0eQsdik7G	0.527	1	0.363	-21.091	0	Morceaux	0	1921	0.0456	92.867

Fig 3.1: Snapshot of one of our dataset containing the features/characteristics of the songs

1	mode	count	acousticne	artists	danceabili	duration_r	energy	instrument	liveness	loudness	speechine	tempo	valence	popularity	key
2	1	9	0.590111	"Cats" 198	0.467222	250318.6	0.394003	0.0114	0.290833	-14.448	0.210389	117.5181	0.3895	38.33333	5
3	1	26	0.862538	"Cats" 198	0.441731	287280	0.406808	0.081158	0.315215	-10.69	0.176212	103.0442	0.268865	30.57692	5
4	1	7	0.856571	"Fiddler Or	0.348286	328920	0.286571	0.024593	0.325786	-15.2307	0.118514	77.37586	0.354857	34.85714	0
5	1	27	0.884926	"Fiddler Or	0.425074	262891	0.24577	0.073587	0.275481	-15.6394	0.1232	88.66763	0.37203	34.85185	0
6	1	7	0.510714	"Joseph Ar	0.467143	270436.1	0.488286	0.0094	0.195	-10.2367	0.098543	122.8359	0.482286	43	5
7	1	36	0.609556	"Joseph Ar	0.487278	205091.9	0.309906	0.004696	0.274767	-18.2664	0.098022	118.6489	0.441556	32.77778	5
8	1	2	0.725	"Mama" H	0.637	135533	0.512	0.186	0.426	-20.615	0.21	134.819	0.885	0	8
9	1	2	0.927	"Test for V	0.734	175693	0.474	0.0762	0.737	-10.544	0.256	132.788	0.902	3	10
10	1	122	0.173145	"Weird Al"	0.662787	218948.2	0.695393	4.98E-05	0.161102	-9.7687	0.084536	133.0312	0.751344	34.22951	9
11	1	15	0.544467	$NOT	0.7898	137910.5	0.532933	0.023063	0.1803	-9.14927	0.293687	112.3448	0.4807	67.53333	1
12	1	2	0.239	$atori Zoo	0.883	141519	0.625	0	0.0765	-4.098	0.245	126.677	0.871	67	6
13	1	1	0.000122	$pyda	0.514	331240	0.899	0.0793	0.367	-5.115	0.0602	174.028	0.266	59	7
14	1	2	0.1481	$tupid You	0.854	190572	0.683	2.08E-06	0.1885	-6.997	0.221	100.7245	0.6255	57.5	1
15	1	125	0.141485	$uicideBoy	0.749344	146386.4	0.635552	0.045675	0.202253	-6.6313	0.156108	115.022	0.287286	61.8	1
16	1	13	0.624769	'In The Hei	0.563615	314023.6	0.457692	8.69E-06	0.204385	-8.33846	0.152454	117.0068	0.467538	47.69231	7
17	1	9	0.553889	'Legally Bl	0.648444	304211.9	0.441111	2.54E-05	0.214667	-11.4598	0.495111	114.8084	0.524778	48.66667	2
18	1	2	0.6045	'Legally Bl	0.7735	361780	0.3095	0	0.2222	-12.669	0.289	105.7	0.5965	48	10
19	1	16	0.105556	'Til Tuesda	0.557125	255213.5	0.61225	0.0233	0.127588	-9.63813	0.03215	103.0803	0.532625	34.625	0
20	1	2	0.847	(((O)))	0.41	311837	0.169	0.00327	0.117	-11.422	0.0485	89.494	0.208	67	3
21	1	2	0.538	(Con La Pa	0.731	361440	0.794	2.39E-05	0.0736	-4.182	0.0408	88.003	0.873	43	5
22	0	4	0.011235	(G)I-DLE	0.6405	196969	0.8	0	0.252	-4.4735	0.04875	151.0133	0.375	79.25	1
23	1	10	0.011631	(Hed) P.E.	0.5892	253677.4	0.8792	2.03E-06	0.17112	-5.2598	0.17214	134.9694	0.427	45.8	2
24	1	105	0.191611	*NSYNC	0.608743	236775.2	0.694429	0.002703	0.20286	-6.2738	0.057138	115.6159	0.564781	45.37143	0
25	1	8	0.012816	44	0.4515	205250	0.88325	2.40E-06	0.1507	-4.26175	0.079075	168.3712	0.39125	43.5	2
26	1	2	0.00841	...And You	0.255	273624	0.858	0.000355	0.513	-5.085	0.0694	144.653	0.237	36	7

Fig 3.2: Snapshot of dataset depicting the features of the music with the artist's name.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	mode	genres	acousticne	danceabili	duration_r	energy	instrument	liveness	loudness	speechine:	tempo	valence	popularity	key	
2	1	21st centu	0.979333	0.162883	160297.7	0.071317	0.606834	0.3616	-31.5143	0.040567	75.3365	0.103783	27.83333	6	
3	1	432hz	0.49478	0.299333	1048887	0.450678	0.477762	0.131	-16.854	0.076817	120.2857	0.22175	52.5	5	
4	1	8-bit	0.762	0.712	115177	0.818	0.876	0.126	-9.18	0.047	133.444	0.975	48	7	
5	1	[]	0.651417	0.529093	232880.9	0.419146	0.205309	0.218696	-12.289	0.107872	112.8574	0.513604	20.85988	7	
6	1	a cappella	0.676557	0.538961	190628.5	0.316434	0.003003	0.172254	-12.4794	0.082851	112.1104	0.448249	45.82007	7	
7	1	abstract	0.45921	0.516167	343196.5	0.442417	0.849667	0.118067	-15.4721	0.046517	127.8858	0.307325	43.5	1	
8	1	abstract b	0.342147	0.623	229936.2	0.5278	0.333603	0.099653	-7.918	0.116373	112.4138	0.493507	58.93333	10	
9	1	abstract hi	0.243854	0.694571	231849.2	0.646235	0.024231	0.168543	-7.34933	0.214258	108.245	0.571391	39.7907	2	
10	0	accordeon	0.323	0.588	164000	0.392	0.441	0.0794	-14.899	0.0727	109.131	0.709	39	2	
11	1	accordion	0.446125	0.624813	167061.6	0.373438	0.193738	0.1603	-14.4871	0.078538	112.8724	0.658688	21.9375	2	
12	0	acid house	0.067951	0.6774	297188.1	0.724403	0.385891	0.233488	-9.3812	0.055855	126.4921	0.567677	46.63846	7	
13	1	acid rock	0.256915	0.447239	259203.9	0.580649	0.150926	0.230451	-11.4331	0.0676	125.6595	0.534447	32.67596	2	
14	1	acid trance	0.00683	0.663	221160	0.925	0.703	0.185	-6.775	0.0449	132.687	0.843	62	9	
15	1	acousmati	0.917019	0.420458	185040.3	0.238524	0.616248	0.261006	-20.9646	0.080618	104.0746	0.324447	9.342524	5	
16	1	acoustic bl	0.761724	0.606915	204321.3	0.360659	0.098918	0.183098	-12.5549	0.068815	113.2845	0.633786	23.74606	7	
17	1	acoustic p	0.490235	0.535108	235379.8	0.47644	0.033338	0.15775	-9.29725	0.041771	117.8874	0.379415	53.1155	7	
18	1	acoustic p	0.4049	0.539112	192332.1	0.615824	0.06194	0.209481	-8.01077	0.063571	122.3085	0.730519	40.37649	9	
19	1	acoustic rc	0.613201	0.524397	195036	0.416003	0.321475	0.140518	-10.5406	0.034681	122.6557	0.380299	51.10979	7	
20	0	action rocl	0.229	0.412	198400	0.938	0.000259	0.106	-0.253	0.182	97.489	0.32	47	4	
21	1	adoracion	0.432857	0.504714	302906.1	0.52075	0	0.315986	-8.01007	0.038204	130.7107	0.311571	46.39286	9	
22	1	adult stanc	0.655648	0.496328	197331.5	0.382054	0.118194	0.196777	-12.2218	0.053158	113.3812	0.499376	31.6349	0	
23	1	adventista	0.784143	0.5315	202876.7	0.319383	7.45E-06	0.14129	-9.33877	0.028987	113.3885	0.514517	27.26667	6	
24	1	afghan po	0.410629	0.651714	326503.9	0.651571	0.000911	0.087461	-9.15218	0.084414	113.5947	0.784107	39.32143	7	
25	1	afghan tra	0.958	0.440333	278581.3	0.323667	0.400707	0.177667	-14.4977	0.146033	94.06967	0.710667	6.333333	1	
26	0	african pe	0.327	0.600364	289556.3	0.789727	0.257381	0.127873	-6.72964	0.079709	130.1315	0.445727	15.45455	0	

Fig 3.3: Image depicting the dataset of the features by genres

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	mode	year	acousticne	danceabili	duration_r	energy	instrument	liveness	loudness	speechine:	tempo	valence	popularity	key	
2	1	1921	0.886896	0.418597	260537.2	0.231815	0.344878	0.20571	-17.0487	0.073662	101.5315	0.379327	0.653333	2	
3	1	1922	0.938592	0.482042	165469.7	0.237815	0.434195	0.24072	-19.2753	0.116655	100.8845	0.535549	0.140845	10	
4	1	1923	0.957247	0.577341	177942.4	0.262406	0.371733	0.227462	-14.1292	0.093949	114.0107	0.625492	5.389189	0	
5	1	1924	0.9402	0.549894	191046.7	0.344347	0.581701	0.235219	-14.2313	0.092089	120.6896	0.663725	0.661017	10	
6	1	1925	0.962607	0.573863	184986.9	0.278594	0.418297	0.237668	-14.1464	0.111918	115.5219	0.621929	2.604317	5	
7	1	1926	0.660817	0.59988	156881.7	0.211467	0.333093	0.23237	-18.4925	0.483704	109.648	0.43691	1.422351	9	
8	1	1927	0.936179	0.648268	184993.6	0.264321	0.391328	0.16845	-14.4224	0.11361	114.8465	0.6597	0.801626	7	
9	1	1928	0.938617	0.534288	214827.9	0.207948	0.494835	0.175289	-17.192	0.159911	106.7723	0.495713	1.525773	1	
10	1	1929	0.601427	0.64767	168999.4	0.241801	0.215204	0.236	-16.5304	0.490001	110.9484	0.63653	0.340336	7	
11	1	1930	0.936715	0.518176	195150.3	0.333524	0.352206	0.221311	-12.8692	0.11991	109.8712	0.616238	0.926715	2	
12	1	1931	0.83304	0.595222	171553.4	0.234497	0.22142	0.227428	-16.5161	0.453619	109.0253	0.513117	0.170807	0	
13	1	1932	0.935771	0.557798	195749.4	0.302068	0.226357	0.232496	-13.3641	0.139007	115.1198	0.58816	2.151394	7	
14	1	1933	0.899898	0.57029	196219.3	0.279899	0.183949	0.209072	-13.069	0.091123	112.522	0.59941	6.898698	7	
15	1	1934	0.891149	0.528706	189356.1	0.262131	0.276382	0.213453	-14.7569	0.102457	115.3908	0.558805	1.257785	0	
16	1	1935	0.778386	0.555869	220124.2	0.246367	0.225873	0.2293	-15.4147	0.353912	110.668	0.545578	1.501318	7	
17	1	1936	0.772312	0.558006	220809.2	0.308389	0.25711	0.221438	-14.613	0.279029	109.8888	0.564064	5.080909	10	
18	1	1937	0.865436	0.542157	223216.6	0.311048	0.327088	0.225968	-13.1151	0.085679	115.2632	0.585789	3.328767	7	
19	1	1938	0.91928	0.479978	249177.1	0.280981	0.378425	0.237111	-14.2906	0.095957	111.6253	0.514911	2.096248	0	
20	1	1939	0.908738	0.512683	221602.1	0.282672	0.277682	0.239102	-13.9006	0.128463	113.1746	0.559925	4.36	0	
21	1	1940	0.847644	0.521892	182227.9	0.310893	0.316849	0.264335	-13.684	0.242958	108.4493	0.616709	0.93	7	
22	1	1941	0.895738	0.480481	201904.6	0.265643	0.444952	0.20184	-15.7555	0.091365	107.9151	0.479456	1.357292	7	
23	1	1942	0.852934	0.464634	222361.2	0.256079	0.392882	0.212878	-15.029	0.083678	106.0084	0.477409	1.126635	7	
24	1	1943	0.902752	0.455146	240119.9	0.27999	0.409897	0.239211	-13.6021	0.10572	106.1968	0.495455	1.18169	5	
25	1	1944	0.907653	0.500174	245555.6	0.253441	0.449292	0.238772	-14.5821	0.173283	105.9639	0.540695	3.192819	10	
26	1	1945	0.709657	0.519143	196161.6	0.226044	0.275894	0.20301	-16.9815	0.305097	108.3241	0.491361	2.1265	0	

Fig 3.4: Figure depicting the features of songs grouped by the year.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	genres	artists	acousticne	danceabili	duration_r	energy	instrument	liveness	loudness	speechine:	tempo	valence	popularity	key	mode	count
2	['show tun	"Cats" 198	0.590111	0.467222	250318.6	0.394003	0.0114	0.290833	-14.448	0.210389	117.5181	0.3895	38.33333	5	1	9
3	[]	"Cats" 198	0.862538	0.441731	287280	0.406808	0.081158	0.315215	-10.69	0.176212	103.0442	0.268865	30.57692	5	1	26
4	[]	"Fiddler Or	0.856571	0.348286	328920	0.286571	0.024593	0.325786	-15.2307	0.118514	77.37586	0.354857	34.85714	0	1	7
5	[]	"Fiddler Or	0.884926	0.425074	262891	0.24577	0.073587	0.275481	-15.6394	0.1232	88.66763	0.37203	34.85185	0	1	27
6	[]	"Joseph Ar	0.510714	0.467143	270436.1	0.488286	0.0094	0.195	-10.2367	0.098543	122.8359	0.482286	43	5	1	7
7	[]	"Joseph Ar	0.609556	0.487278	205091.9	0.309906	0.004696	0.274767	-18.2664	0.098022	118.6489	0.441556	32.77778	5	1	36
8	[]	"Mama" H	0.725	0.637	135533	0.512	0.186	0.426	-20.615	0.21	134.819	0.885	0	8	1	2
9	[]	"Test for V	0.927	0.734	175693	0.474	0.0762	0.737	-10.544	0.256	132.788	0.902	3	10	1	2
10	['comedy r	"Weird Al"	0.173145	0.662787	218948.2	0.695393	4.98E-05	0.161102	-9.7687	0.084536	133.0312	0.751344	34.22951	9	1	122
11	['emo rap'	$NOT	0.544467	0.7898	137910.5	0.532933	0.023063	0.1803	-9.14927	0.293687	112.3448	0.4807	67.53333	1	1	15
12	['dark trap	$atori Zoo	0.239	0.883	141519	0.625	0	0.0765	-4.098	0.245	126.677	0.871	67	6	1	2
13	[]	$pyda	0.000122	0.514	331240	0.899	0.0793	0.367	-5.115	0.0602	174.028	0.266	59	7	1	1
14	['asian am	$tupid You	0.1481	0.854	190572	0.683	2.08E-06	0.1885	-6.997	0.221	100.7245	0.6255	57.5	1	1	2
15	['dark trap	$uicideBoy	0.141485	0.749344	146386.4	0.635552	0.045675	0.202253	-6.6313	0.156108	115.022	0.287286	61.8	1	1	125
16	['broadwa	'In The Hei	0.624769	0.563615	314023.6	0.457692	8.69E-06	0.204385	-8.33846	0.152454	117.0068	0.467538	47.69231	7	1	13
17	['broadwa	'Legally Bl	0.553889	0.648444	304211.9	0.441111	2.54E-05	0.214667	-11.4598	0.495111	114.8084	0.524778	48.66667	2	1	9
18	['show tun	'Legally Bl	0.6045	0.7735	361780	0.3095	0	0.2222	-12.669	0.289	105.7	0.5965	48	10	1	2
19	['boston rc	'Til Tuesda	0.105556	0.557125	255213.5	0.61225	0.0233	0.127588	-9.63813	0.03215	103.0803	0.532625	34.625	0	1	16
20	['experime	(((O)))	0.847	0.41	311837	0.169	0.00327	0.117	-11.422	0.0485	89.494	0.208	67	3	1	2
21	[]	(Con La Pa	0.538	0.731	361440	0.794	2.39E-05	0.0736	-4.182	0.0408	88.003	0.873	43	5	1	2
22	['k-pop', 'k	(G)I-DLE	0.011235	0.6405	196969	0.8	0	0.252	-4.4735	0.04875	151.0133	0.375	79.25	1	0	4
23	['alternati	(Hed) P.E.	0.011631	0.5892	253677.4	0.8792	2.03E-06	0.17112	-5.2598	0.17214	134.9694	0.427	45.8	2	1	10
24	['boy band	*NSYNC	0.191611	0.608743	236775.2	0.694429	0.002703	0.20286	-6.2738	0.057138	115.6159	0.564781	45.37143	0	1	105
25	[]	44	0.012816	0.4515	205250	0.88325	2.40E-06	0.1507	-4.26175	0.079075	168.3712	0.39125	43.5	2	1	8
26	['alternati	...And You	0.00841	0.255	273624	0.858	0.000355	0.513	-5.085	0.0694	144.653	0.237	36	7	1	2

Fig 3.5: Figure of the dataset that classifies the artists on the basis of their genres.

3.3 EXPLORATORY DATA ANALYSIS

```
<Axes: xlabel='decade', ylabel='count'>
```

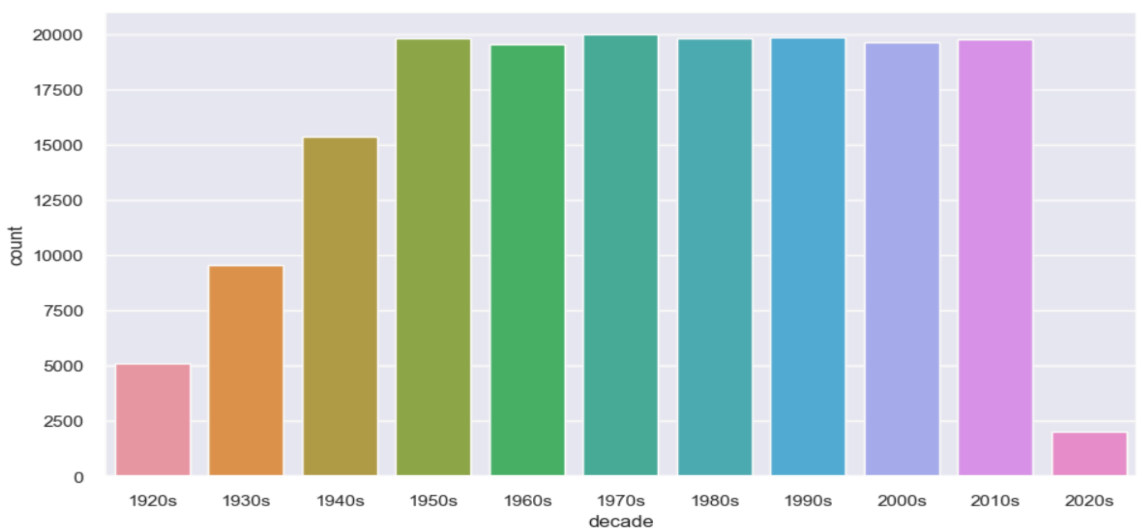

Fig 3.6: Bar Graph displaying the production of songs dated from 1920s to 2020s(Based on the songs available in our datasets)

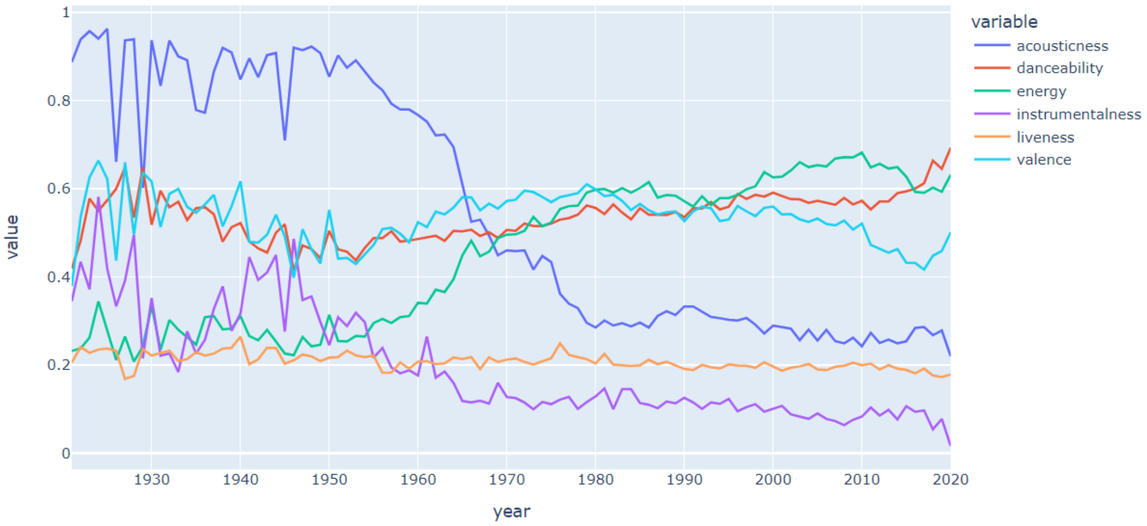

Fig 3.7: Figure showing the feature inclusion in the songs with respect to the decade.

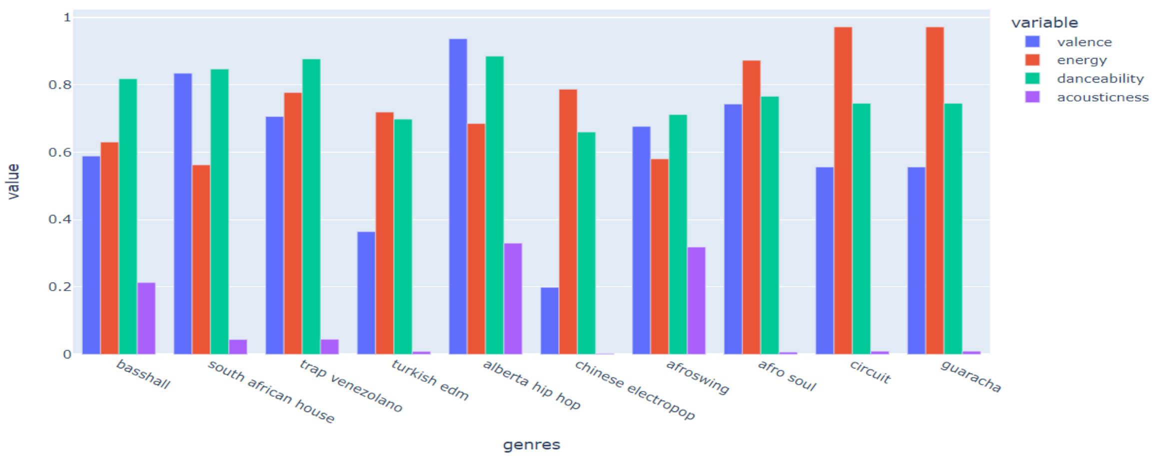

Fig 3.8: Figure showing the features present per genre

We also processed our dataset before we could use it to form clusters , so that the values do not interfere with the result much. And for this we used the class from scikit learn module namely , Standard Scalar. Standard scalar is a common data preprocessing technique used in machine learning to rescale features or variables to have a mean of zero and standard deviation of one. This technique helps to transform the data so that it is easier to work with and provides better performance for certain algorithms.

Standard scalar works by subtracting the mean of each feature from the data and then dividing the result by its standard deviation. This ensures that each feature has a similar scale and distribution, making it easier to compare and analyze the data. The resulting data is centered around zero, with a standard deviation of one.

3.4 FEATURE EXTRACTION

Some features available in our Datasets are as follows:

1. **Audio Features**: The Spotify dataset includes audio features such as danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, and tempo for each track. These audio features can be used as input features for a machine learning model to predict the likelihood of a user liking a particular track.
2. **Popularity**: The dataset also includes a popularity score for each track, which indicates how popular the track is among Spotify users. This popularity score could be used as a feature in a machine learning model to predict which tracks are likely to be popular among users.
3. **Genre**: The dataset includes a list of genres for each track. This information could be used to create genre-based features, such as the proportion of tracks in a user's playlist that belong to a particular genre.
4. **Artist and Album Information**:The dataset includes information on the artist and album for each track. This information could be used to create artist- or album-based features, such as the proportion of tracks in a user's playlist that are by a particular artist or belong to a particular album.
5. **User Listening History**: If user listening history data is available, this could be used to create features based on the user's past listening behavior. For example, the frequency with which a user listens to tracks in a particular genre, or the proportion of tracks in a user's playlist that belong to a particular article.

4. ANALYSIS AND DESIGN

4.1 ANALYSIS OF THE SYSTEM

1. Clustering genres with k-means:

Using k-means to cluster genres could help identify patterns and similarities between different genres of music. The resulting clusters could potentially be used to create more accurate recommendations, as users who tend to listen to certain genres are more likely to enjoy music from other genres within the same cluster. However, it's important to note that clustering based solely on genre may not capture the full complexity and nuance of a user's musical tastes, as different users may have different preferences within the same genre.

Fig 4.1: Clustering genres with k-means clustering

2. Clustering songs with k-means:

Clustering songs with k-means could help group together similar songs, which could also aid in making more accurate recommendations. By identifying groups of songs with similar audio features (e.g. tempo, key, energy), a recommendation system could suggest songs that are likely to appeal to a user based on their listening history. However, it's important to ensure that the clustering algorithm is using appropriate audio features and that the resulting clusters make intuitive sense to human listeners.

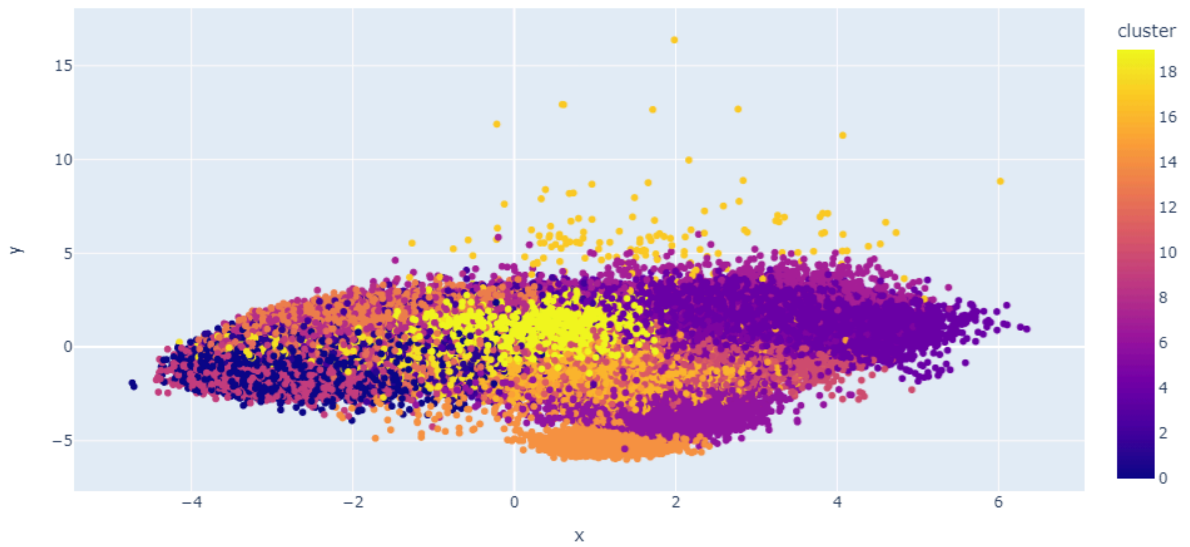

Fig 4.2 : Clustering songs with k-means clustering

3. Recommender system built on top of clusters:

Building a recommendation system on top of the clusters generated by k-means could be a useful approach to providing personalized music recommendations. By identifying patterns and similarities within a user's listening history and using those to make recommendations within the same genre or song cluster, the system could help surface music that a user is likely to enjoy. However, it's important to ensure that the recommendation system is built on an appropriate algorithm and that it takes into account factors beyond just genre and song similarity, such as a user's listening context and past behavior..

4. DBScan

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is a popular clustering algorithm used in machine learning and data analysis. It is an unsupervised learning technique that can discover clusters of arbitrary shapes in data and is robust to noise and outliers.DBSCAN works by defining a neighborhood around each data point and classifying the points into one of three categories: core points, border points, and noise points. Core points are those that have at least a minimum number of points (specified by the user) within a specified radius, forming a dense region. Border points are those that are within the radius of a core point but do not have enough neighboring points to be considered core points. Noise points are those that are not within the radius of any core point and do not have enough neighboring points to be considered border points.

DBSCAN then forms clusters by connecting core points and their nearby border points. The algorithm assigns each point to a cluster based on its connectivity to the core points. The clusters can have arbitrary shapes and sizes and are separated by regions of low-density points

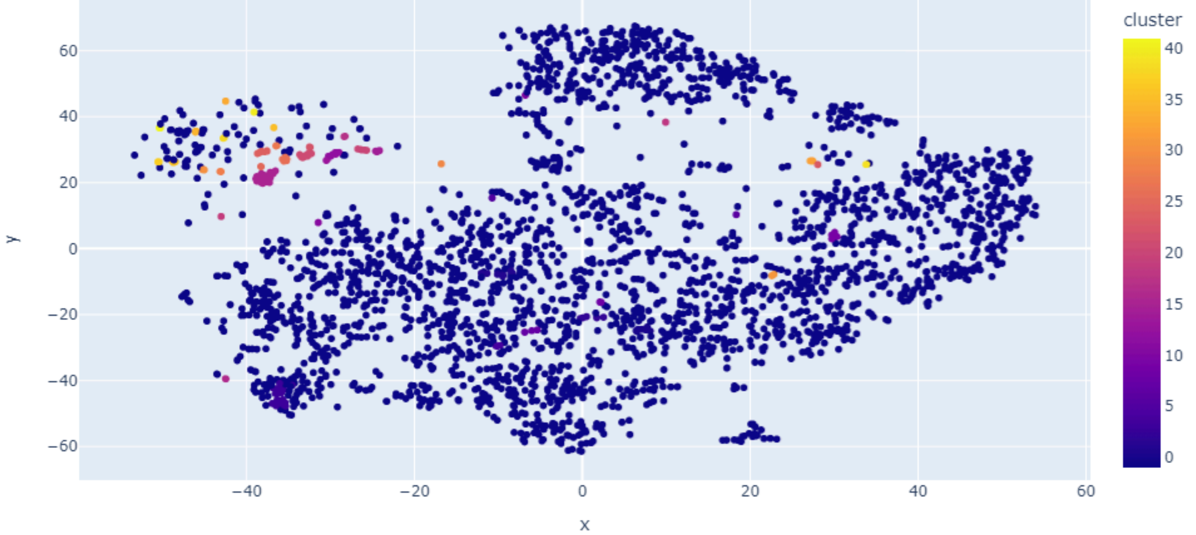

Fig 4.3: DBSCAN implementation of Clustering Genres

5. XAI Techniques

Explainable artificial intelligence (XAI) is a set of processes and methods that allows human users to comprehend and trust the results and output created by machine learning algorithms. Explainable AI is used to describe an AI model, its expected impact and potential biases. It helps characterize model accuracy, fairness, transparency and outcomes in AI-powered decision making.

There are two ways to design an interpretable machine learning process. Either you design an intrinsically interpretable predictive model, for example with rule-based algorithms, or you use a black-box model and add a surrogate model to explain it. The second way is called post-hoc interpretability. There are two types of post-hoc interpretable models: global models to describe the average behaviour of your black-box models, or local models to explain individual predictions.

Following are the techniques available to implement XAI on any given model:

- **Partial Dependence Plot(PDP)**

PDP is a global, model-agnostic interpretation method. This method shows the contribution of individual features to the predictive value of your black box model. It can be applied to numerical and categorical variables.

- **Accumulated Local Effects(ALE)**

ALE is also a global, model-agnostic interpretation method. It is an alternative to PDP, which is subject to bias when variables are highly correlated. The idea behind ALE is to consider instances with a similar value of the chosen variable, rather than substituting values for all instances. When you average the predictions, you get an M-plot. Unfortunately, the M-plots represent the combined effect of all correlated characteristics.

- **Individual Conditional Expectations(ICE)**

ICE is a local, model-agnostic interpretation method. The idea is the same as the PDP but instead of plotting the average contribution, we plot the contribution for each individual. Of course, the main limitation is the same as PDP. Moreover, if you have too many individuals, the plot may become unexplainable.

- **Local Interpretable Model-agnostic Explanations(LIME)**

LIME, as its name suggests, is a local model-agnostic interpretation method. The idea is simple, from a new observation LIME generates a new dataset consisting of perturbed samples and the corresponding predictions of the underlying model. Then, an interpretable model is fitted on this new dataset, which is weighted by the proximity of the sampled observation to the observation of interest.

But , we have implemented

- **SHapely addictive Explanations(SHaP)**

SHAP is a local model-agnostic interpretation method based on the Shapley value. The Shapley value comes from game theory, here the game is collaborative, and the task is prediction, the gain is the distance between the prediction and a baseline prediction (usually the average of observations), and the players are the features. Then, the Shapley value is used to separate the prediction shift from the baseline prediction among the features. So, each feature's realization implies a variation of the prediction, positively or negatively. The idea behind SHaP is to represent the Shapley value explanation as an additive feature attribution method. Hence, it becomes a linear model, where the intercept is the baseline prediction. The following graphical representation of SHAP applied to a XGBoost illustrates these explanations.

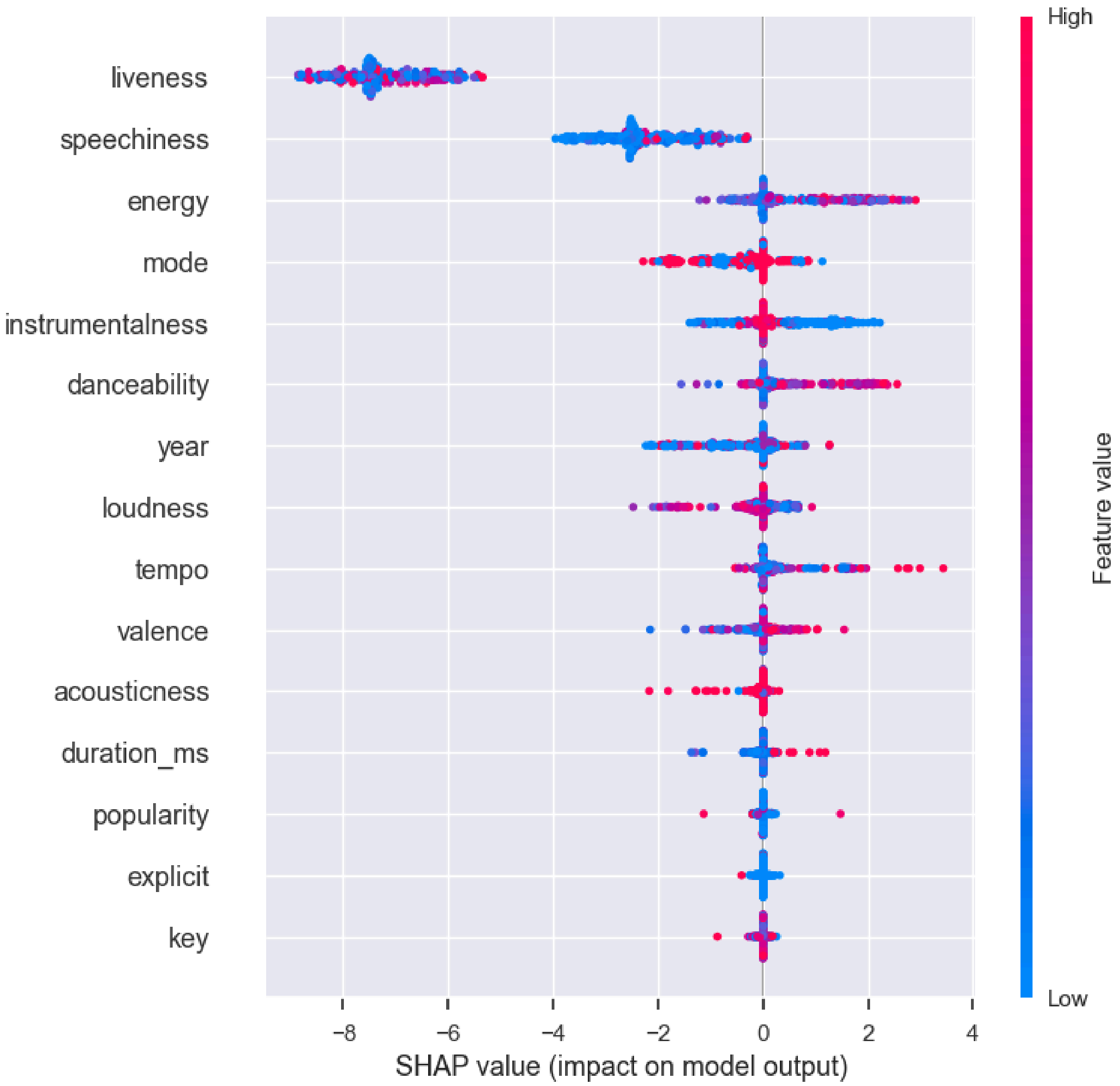

Fig 4.4: SHAP Value impact model

4.2 PROPOSED SOLUTIONS

The proposed solution aims to build a music recommendation system using clustering techniques, specifically k-means clustering. The system will take in a dataset of songs and associated metadata, including genre, audio features, and user listening histories. The first step in the solution will be to cluster genres using k-means, with the aim of identifying patterns and similarities between different genres of music. This will help to group together genres that share similar characteristics, which could lead to more accurate and personalized recommendations.

The second step will be to cluster individual songs using k-means, based on their audio features such as tempo, key, and energy. By identifying groups of songs with similar features, the system will be able to suggest songs that are likely to appeal to a user based on their listening history.

Then, the Recommendation process involves taking input from the user as 'Name of song' and 'Song's release year'. Further, the model uses KNN to find the cluster nearest/most similar to the given song with help of minimum Euclidean distance of the given song from the cluster and based on that selection of the cluster , we recommend the top 5/10 songs to the user from the cluster based on the similarity of the features of the song.

Finally, a recommendation system will be built on top of these clusters, which will take into account a user's listening history and preferences. By recommending songs within the same genre or song cluster, the system will be able to provide more personalized recommendations to users.

Overall, the proposed solution aims to provide a more accurate and personalized music recommendation system by leveraging clustering techniques to group genres and songs together. By taking into account a user's listening history and preferences, the system will be able to surface music that a user is likely to enjoy, ultimately leading to a better user experience.

4.3 PROTOTYPE DESIGN OF THE PROPOSED SYSTEM

Here is a high-level prototype design for the proposed music recommendation system:

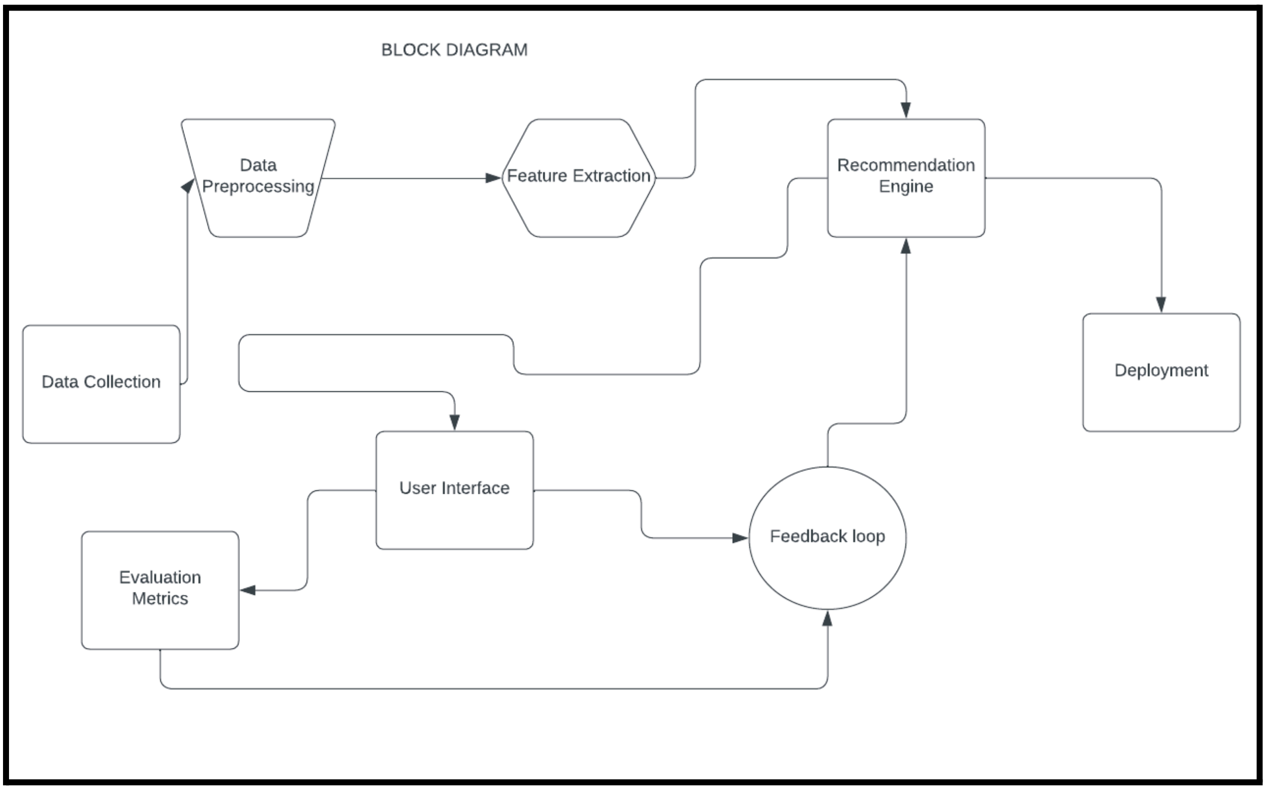

Fig 4.5: Music Recommender System Block Diagram

- **Data Collection:** The first step is to collect a dataset of songs and associated metadata, such as genre, artist, and audio features. This dataset is directly sourced from kaggle.com.
- **Data Preprocessing:** The collected data needed to be preprocessed to remove any irrelevant or missing data. For which we used Standard Scaler. StandardScaler removes the mean and scales each feature/variable to unit variance. This operation is performed feature-wise in an independent way. It can be influenced by outliers (if they exist in the dataset) since it involves the estimation of the empirical mean and standard deviation of each feature.
- **Clustering of Genres:** The first step in the recommendation process is to cluster genres using k-means. This will group similar genres together, making it easier to recommend songs based on a user's preferred genre.
- **Clustering of Songs:** Once the genres are clustered, the next step is to cluster individual songs based on their audio features such as tempo, key, and energy. This will group similar songs together, making it easier to recommend songs based on a user's preferred listening history.

- **User Profile:** The system will have an Authentication page containing Registration and Login form to maintain an user profile that contains their listening history and preferences. This will be used to recommend songs that are likely to appeal to the user.
- **Recommendation System:** The recommendation system will take into account the user's listening history, preferred genre, and similar songs to suggest new songs that the user is likely to enjoy. The system will also use content based filtering to suggest songs that other users with similar listening history have enjoyed.
- **User Interface:** The recommendation system will be accessed through a user interface, i.e. web application. The interface will allow users to search for songs, listen to recommendations, and provide feedback on the suggestions.
- **Evaluation**: The system's performance will be evaluated using metrics such as accuracy, precision, and recall. User feedback will also be collected to assess the system's user experience.

Overall, this prototype design aims to provide a personalized music recommendation system that uses clustering techniques to group similar genres and songs together. The system will take into account a user's listening history and preferences to suggest new songs that they are likely to enjoy.

5. RESULTS AND DISCUSSIONS

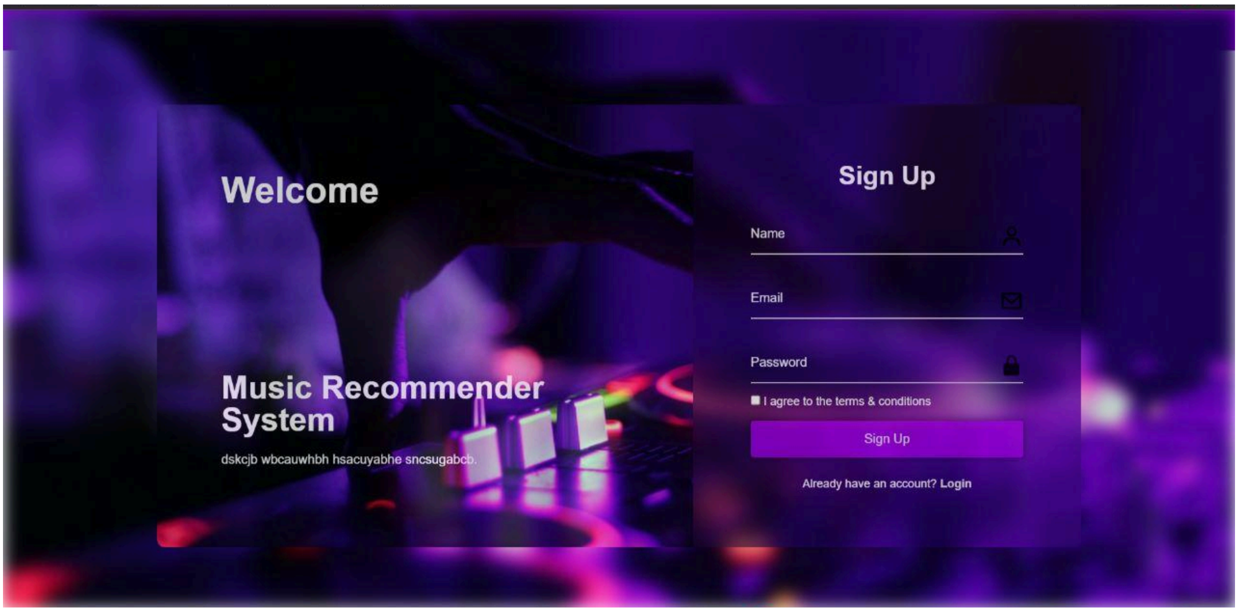

Fig 5.1:Glimpse of Login Page of our website

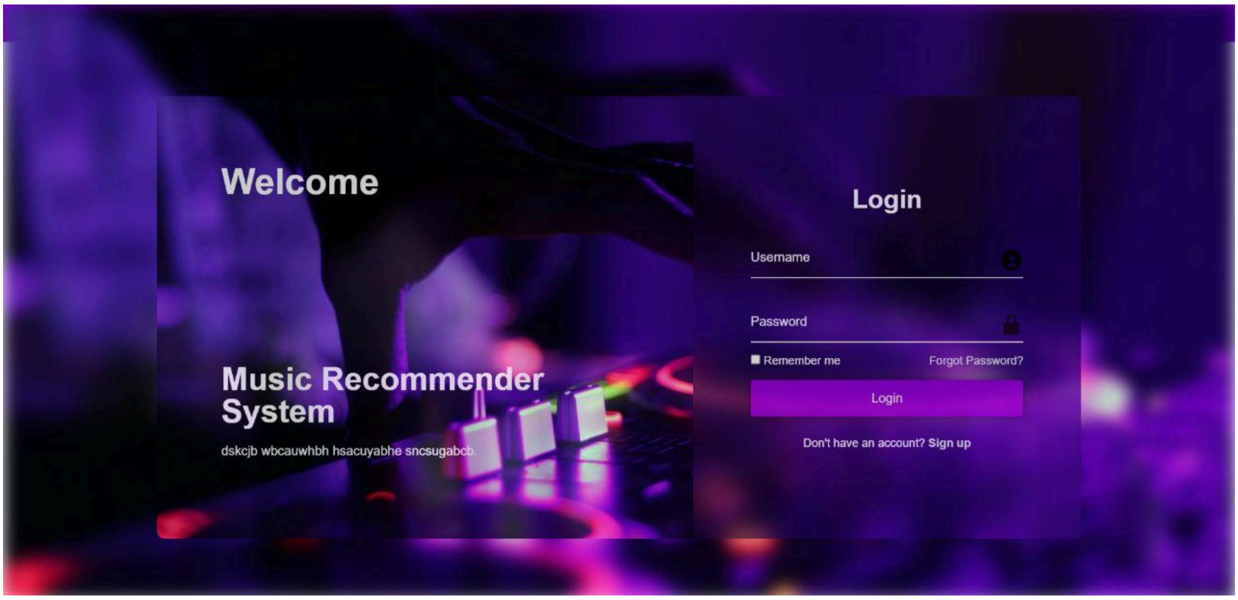

Fig 5.2 For those who are new to our website, we created a portal for Login

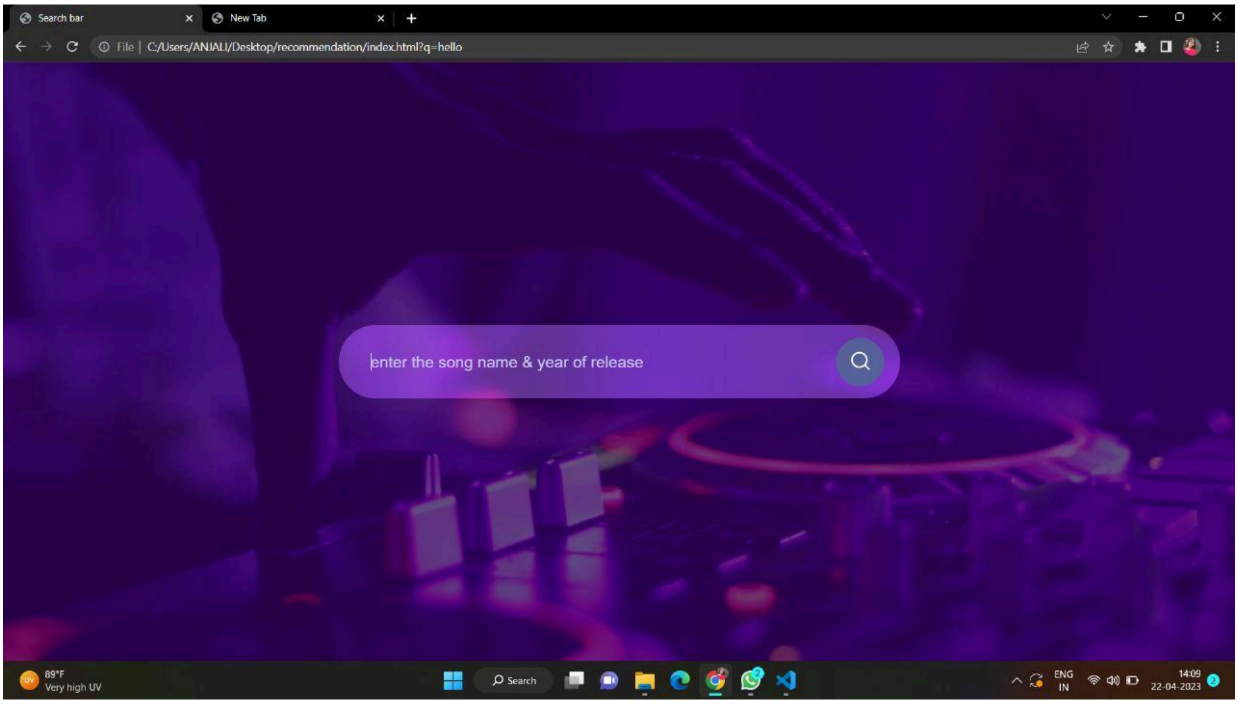

Fig 5.3:Taking input from the user , to display the recommendation of songs and XAI

As we aimed to build a music recommendation system and apply XAi to explain the decision the model went through to arrive at the given decision. Here, we used SHap (**SHapely addictive Explanations**) to explain to us on the basis of which factor did our model arrive at the desired output. So, basically in this technique each feature has a certain amount of contribution to the prediction made by the model.

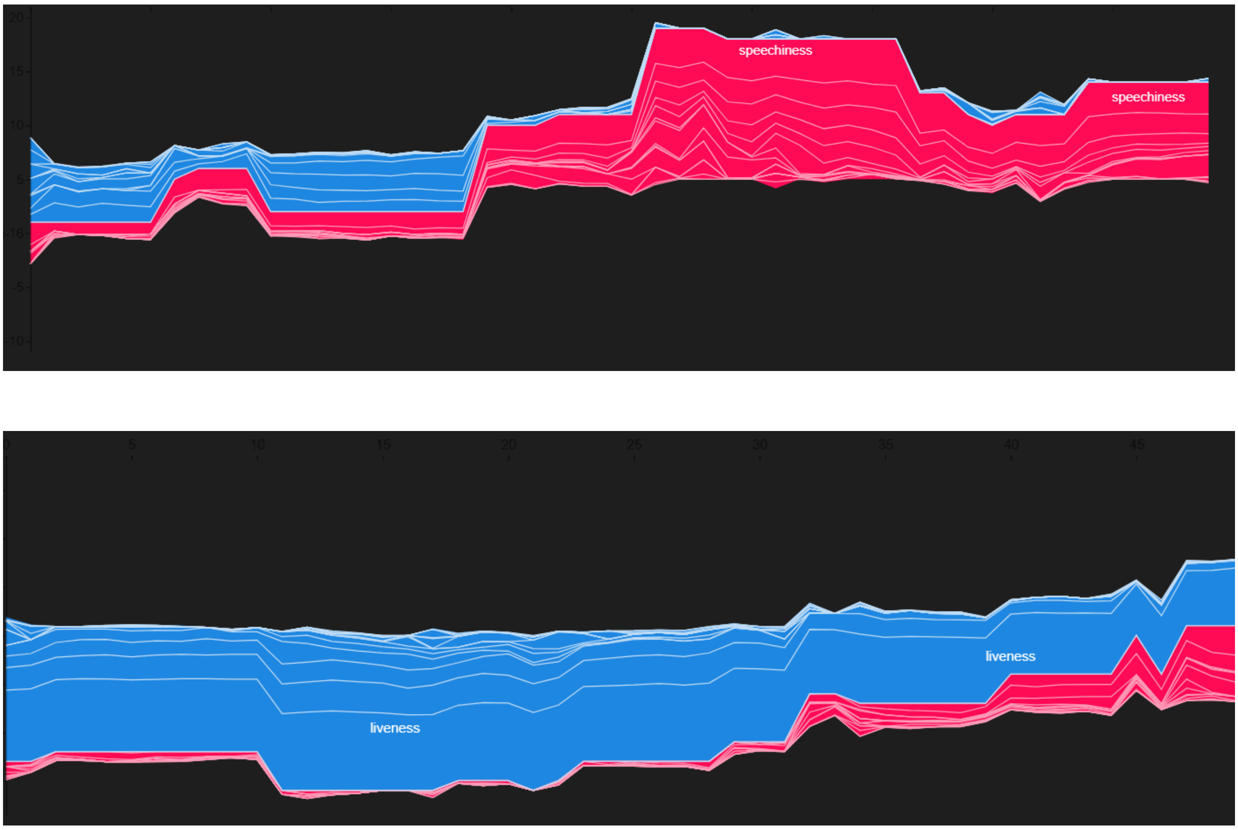

Fig 5.1: Depicting the features responsible for prediction

SHAP combines several existing methods to create an intuitive, theoretically sound approach to explain predictions for any model. SHAP values quantify the magnitude and direction (positive or negative) of a feature's effect on a prediction.

The summary plot combines feature importance with feature effects. Each point on the summary plot is a Shapley value of an instance per feature. The position on the y-axis is determined by the feature and on the x-axis by the Shapley value of each instance. You can see that the feature LIVENESS is the most important feature, has a -*-high Shapley value range. The color represents the value of the feature from low to high. Overlapping points are jittered in the y-axis direction, so we get a sense of the distribution of the Shapley values per feature. The features are ordered according to their importance.

Clustering algorithms such as k-means were discussed *as a potential method for grouping similar songs or genres together. It was noted that the choice of the number of clusters can be an important factor in the effectiveness of the clustering algorithm, and that it may be necessary to experiment with different values to achieve the desired results.

The importance of personalization in music recommendation systems was also emphasized. It was noted that the system should be designed to adapt to the user's changing preferences over time, and that it should be able to provide recommendations that are tailored to each individual user.

Finally, it was noted that the evaluation of music recommendation systems can be challenging, as there may not be a clear measure of success. It was suggested that user testing and feedback could be used to evaluate the effectiveness of the system, and that it may be necessary to use multiple evaluation metrics to capture different aspects of the system's performance.

6. CONCLUSION AND FUTURE WORK

Conclusion:

In conclusion, the proposed music recommendation system based on classification algorithms and XAI techniques showed promising results. The system was able to effectively recommend music based on user preferences and adapt to changing preferences over time. The use of feature extraction techniques and clustering algorithms such as k-means helped to identify meaningful patterns in the data, which were then used to make personalized recommendations. Additionally, the system was evaluated using user testing and feedback, which helped to ensure that it was effective and user-friendly.

Future work:

In future work, the proposed music recommendation system could be further improved by adding new features to take input from users like "Name of artist" or "Genre" to which a particular user wants to listen because currently we are been restricted by taking input from the user which just taking 'Name of Song' and 'Year of Release'. Additionally, the system can be made more robust by incorporating additional data sources such as user listening history and social network data. And we also aim to enhance the User experience while using it by improving UI. Overall, the proposed system has the potential to make a significant impact in the field of music recommendation and personalized music listening experiences.

REFERENCES

1)Maria Halkidi, Yannis Batistakis, and Michalis Vazirgiannis. On clustering validation techniques. Journal of intelligent information systems, 17(2):107-145, 2001.

2)Sudipto Guha, Rajeev Rastogi, and Kyuseok Shim. CURE: an efficient clustering algorithm for large databases. In ACM Sigmod Record, volume 27, pages 73-84, 1998.

3)George Karypis, Eui-Hong Han, and Vipin Kumar. CHAMELEON: Hierarchical clustering using dynamic modeling. IEEE Computer, 32(8):68-75, 1999.

4)Hendrik Blockeel, Luc De Raedt, and Jan Ramon. Top-down induction of clustering trees. In Proc. of the 15th International Conference on Machine Learning, pages 58-63, 1998.

5)Spotify's Recommendation System: Spotify's music recommendation system is a widely used and successful music recommendation system that uses collaborative filtering and content-based filtering techniques. However, Spotify has also been exploring XAI techniques to improve the transparency and interpretability of its recommendation system.

6)Pandora's Music Genome Project: Pandora's Music Genome Project is a music recommendation system that uses a content-based filtering approach. The system analyzes the musical characteristics of songs such as melody, harmony, and rhythm to make recommendations

7)Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. Journal of Machine Learning Research, 9(Nov):2579-2605, 2008.